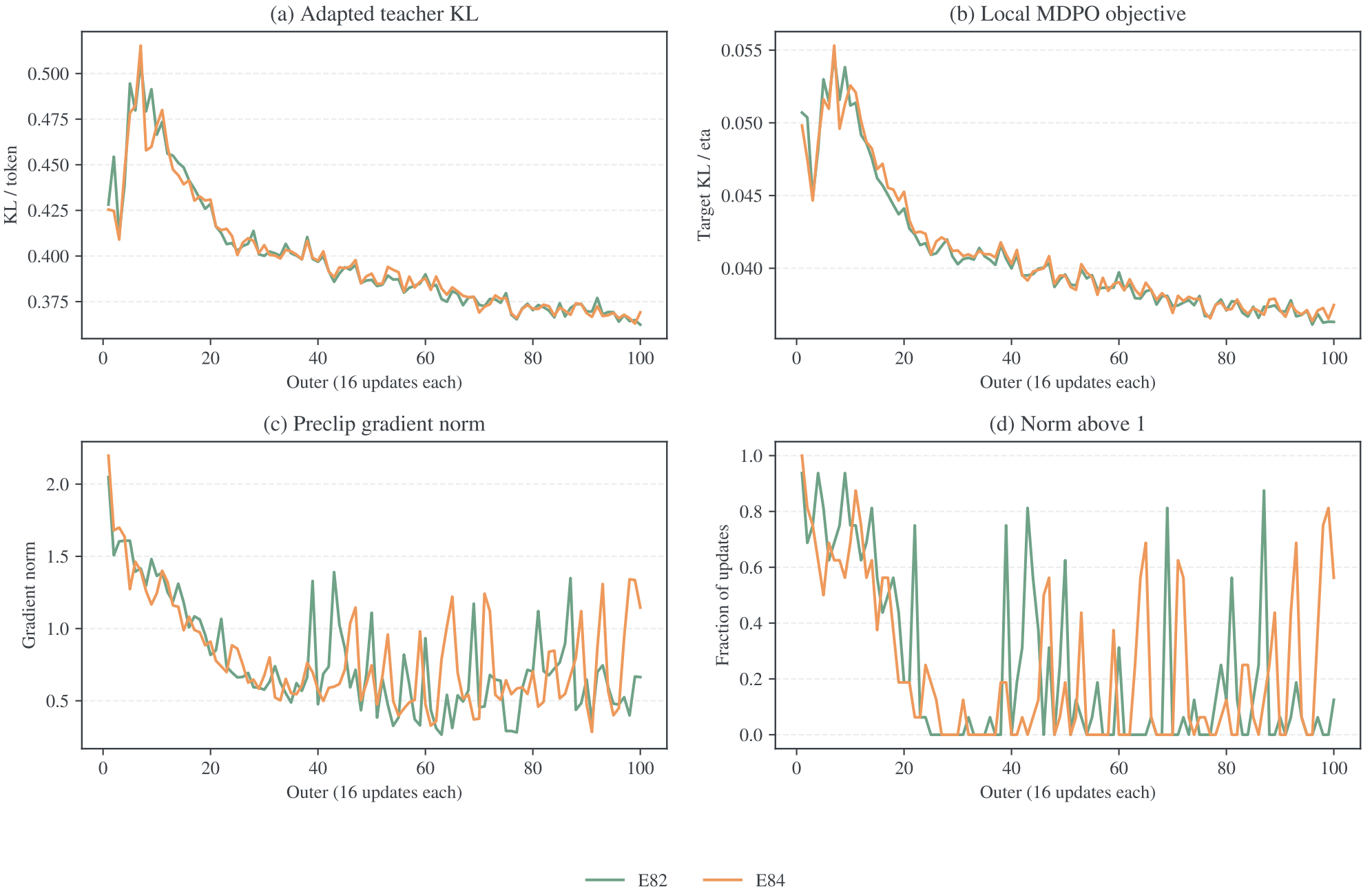
(a) Adapted teacher KL
KL / token
0.500
0.475
0.450
0.425
0.400
0.375
0
20
40
60
80
100
Outer (16 updates each)
(b) Local MDPO objective
Target KL / eta
0.055
0.050
0.045
0.040
0
20
40
60
80
100
Outer (16 updates each)
(c) Preclip gradient norm
Gradient norm
2.0
1.5
1.0
0.5
0
20
40
60
80
100
Outer (16 updates each)
(d) Norm above 1
Fraction of updates
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Outer (16 updates each)
E82
E84
Block means on each run's own trajectories; same teacher-KL metric, changing visited prefixes